

```
In [1]: import pandas as pd
import os
```

```
In [2]: pip install duckdb
```

Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: duckdb in c:\users\703366897\appdata\roaming\python\python312\site-packages (1.5.5)
Note: you may need to restart the kernel to use updated packages.
[notice] A new release of pip is available: 24.0 -> 26.2.1
[notice] To update, run: python.exe -m pip install --upgrade pip

```
In [3]: import duckdb
print(duckdb.__version__)
```

1.5.5

```
In [4]: con = duckdb.connect()

con.sql("""
SELECT 'Hello SQL Project' AS message
""").show()
```

message varchar
Hello SQL Project

```
In [5]: customers = pd.read_csv(
    "olist_SQL_Project/data/olist_customers_dataset.csv"
)

customers.head()
```

```
Out[5]:
```

	customer_id	customer_unique_id	customer_zip_cc
0	06b8999e2fba1a1fbc88172c00ba8bc7	861eff4711a542e4b93843c6dd7febb0	
1	18955e83d337fd6b2def6b18a428ac77	290c77bc529b7ac935b93aa66c333dc3	
2	4e7b3e00288586ebd08712fdd0374a03	060e732b5b29e8181a18229c7b0b2b5e	
3	b2b6027bc5c5109e529d4dc6358b12c3	259dac757896d24d7702b9acbbff3f3c	
4	4f2d8ab171c80ec8364f7c12e35b23ad	345ecd01c38d18a9036ed96c73b8d066	

```
In [6]: orders = pd.read_csv(
    "olist_SQL_Project/data/olist_orders_dataset.csv"
)
```

```
orders.head()
```

Out[6]:

	order_id	customer_id	order_status	on
0	e481f51cbdc54678b7cc49136f2d6af7	9ef432eb6251297304e76186b10a928d	delivered	
1	53cdb2fc8bc7dce0b6741e2150273451	b0830fb4747a6c6d20dea0b8c802d7ef	delivered	
2	47770eb9100c2d0c44946d9cf07ec65d	41ce2a54c0b03bf3443c3d931a367089	delivered	
3	949d5b44dbf5de918fe9c16f97b45f8a	f88197465ea7920adcdbec7375364d82	delivered	
4	ad21c59c0840e6cb83a9ceb5573f8159	8ab97904e6daea8866dbdbc4fb7aad2c	delivered	

In [7]:

```
order_items = pd.read_csv(  
    "olist_SQL_Project/data/olist_order_items_dataset.csv"  
)  
order_items.head()
```

Out[7]:

	order_id	order_item_id	product_id
0	00010242fe8c5a6d1ba2dd792cb16214	1	4244733e06e7ecb4970a6e2683c13e61
1	00018f77f2f0320c557190d7a144bdd3	1	e5f2d52b802189ee658865ca93d83a8f
2	000229ec398224ef6ca0657da4fc703e	1	c777355d18b72b67abbeef9df44fd0fd
3	00024acbcdf0a6daa1e931b038114c75	1	7634da152a4610f1595efa32f14722fc
4	00042b26cf59d7ce69dfabb4e55b4fd9	1	ac6c3623068f30de03045865e4e10089

In [8]:

```
products = pd.read_csv("olist_SQL_Project/data/olist_products_dataset.csv")  
products.head()
```

Out[8]:

	product_id	product_category_name	product_name_lenght	prod
0	1e9e8ef04dbcff4541ed26657ea517e5	perfumaria	40.0	
1	3aa071139cb16b67ca9e5dea641aaa2f	artes	44.0	
2	96bd76ec8810374ed1b65e291975717f	esporte_lazer	46.0	
3	cef67bcfe19066a932b7673e239eb23d	bebes	27.0	
4	9dc1a7de274444849c219cff195d0b71	utilidades_domesticas	37.0	

```
In [9]: sellers = pd.read_csv("Olist_SQL_Project/data/olist_sellers_dataset.csv")
sellers.head()
```

```
Out[9]:
```

	seller_id	seller_zip_code_prefix	seller_city	seller_state
0	3442f8959a84dea7ee197c632cb2df15	13023	campinas	SP
1	d1b65fc7debc3361ea86b5f14c68d2e2	13844	mogi guacu	SP
2	ce3ad9de960102d0677a81f5d0bb7b2d	20031	rio de janeiro	RJ
3	c0f3eea2e14555b6faeea3dd58c1b1c3	4195	sao paulo	SP
4	51a04a8a6bdcdb23deccc82b0b80742cf	12914	braganca paulista	SP

```
In [10]: payments = pd.read_csv("Olist_SQL_Project/data/olist_order_payments_dataset.csv")
payments.head()
```

```
Out[10]:
```

	order_id	payment_sequential	payment_type	payment_installments
0	b81ef226f3fe1789b1e8b2acac839d17	1	credit_card	
1	a9810da82917af2d9aefd1278f1dcfa0	1	credit_card	
2	25e8ea4e93396b6fa0d3dd708e76c1bd	1	credit_card	
3	ba78997921bbcdc1373bb41e913ab953	1	credit_card	
4	42fdf880ba16b47b59251dd489d4441a	1	credit_card	

```
In [11]: reviews = pd.read_csv("Olist_SQL_Project/data/olist_order_reviews_dataset.csv")
reviews.head()
```

```
Out[11]:
```

	review_id	order_id	review_score	review_text
0	7bc2406110b926393aa56f80a40eba40	73fc7af87114b39712e6da79b0a377eb	4	
1	80e641a11e56f04c1ad469d5645fdfde	a548910a1c6147796b98fdf73dbeba33	5	
2	228ce5500dc1d8e020d8d1322874b6f0	f9e4b658b201a9f2ecdecbb34bed034b	5	
3	e64fb393e7b32834bb789ff8bb30750e	658677c97b385a9be170737859d3511b	5	
4	f7c4243c7fe1938f181bec41a392bdeb	8e6bfb81e283fa7e4f11123a3fb894f1	5	

```
In [12]: translation = pd.read_csv("Olist_SQL_Project/data/product_category_name_translation.csv")
translation.head()
```

```
Out[12]:
```

	product_category_name	product_category_name_english
0	beleza_saude	health_beauty
1	informatica_acessorios	computers_accessories
2	automotivo	auto
3	cama_mesa_banho	bed_bath_table
4	moveis_decoracao	furniture_decor

```
In [13]: geolocation = pd.read_csv("Olist_SQL_Project/data/olist_geolocation_dataset.csv")
geolocation.head()
```

```
Out[13]:
```

	geolocation_zip_code_prefix	geolocation_lat	geolocation_lng	geolocation_city	geolocati
0	1037	-23.545621	-46.639292	sao paulo	
1	1046	-23.546081	-46.644820	sao paulo	
2	1046	-23.546129	-46.642951	sao paulo	
3	1041	-23.544392	-46.639499	sao paulo	
4	1035	-23.541578	-46.641607	sao paulo	

```
In [14]: customers.shape
```

```
Out[14]: (99441, 5)
```

```
In [15]: customers.columns
```

```
Out[15]: Index(['customer_id', 'customer_unique_id', 'customer_zip_code_prefix',
               'customer_city', 'customer_state'],
              dtype='str')
```

```
In [16]: customers.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 99441 entries, 0 to 99440
Data columns (total 5 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   customer_id                          99441 non-null  str
1   customer_unique_id                   99441 non-null  str
2   customer_zip_code_prefix             99441 non-null  int64
3   customer_city                        99441 non-null  str
4   customer_state                       99441 non-null  str
dtypes: int64(1), str(4)
memory usage: 3.8 MB
```

```
In [17]: customers.isnull().sum()
```

```
Out[17]: customer_id          0
customer_unique_id         0
customer_zip_code_prefix    0
customer_city               0
customer_state              0
dtype: int64
```

```
In [18]: customers.sample(5)
```

```
Out[18]:
```

	customer_id	customer_unique_id	customer_zi
48272	1676edf55749e467544f15922750a231	7a735b500033f8e5957f28a80a3d6fdf	
23258	342e317a94e99f9100ce447a1ea53fb1	1d441954acebde8f2713904082a1c9d7	
74078	dd5fe772009b97031b9e497278667f5c	e7445ffa291c70f6e4e68379acc9ee7a	
5493	654462019c31f990e4e83b074b585b0e	0aab2a678227c8ea1cab4315e5c1094a	
75677	926a39c05e638211462221c33ae1fa5a	495da79af2388e1609849b52d6d4a348	

```
In [19]: orders.shape
```

```
Out[19]: (99441, 8)
```

```
In [20]: orders.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 99441 entries, 0 to 99440
Data columns (total 8 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   order_id                             99441 non-null  str
1   customer_id                          99441 non-null  str
2   order_status                         99441 non-null  str
3   order_purchase_timestamp             99441 non-null  str
4   order_approved_at                    99281 non-null  str
5   order_delivered_carrier_date         97658 non-null  str
6   order_delivered_customer_date        96476 non-null  str
7   order_estimated_delivery_date        99441 non-null  str
dtypes: str(8)
memory usage: 6.1 MB
```

```
In [21]: orders.isnull().sum()
```

```
Out[21]: order_id          0
customer_id          0
order_status         0
order_purchase_timestamp  0
order_approved_at     160
order_delivered_carrier_date 1783
order_delivered_customer_date 2965
order_estimated_delivery_date  0
dtype: int64
```

```
In [22]: customers['customer_unique_id'].nunique()
```

```
Out[22]: 96096
```

```
In [23]: customers['customer_id'].nunique()
```

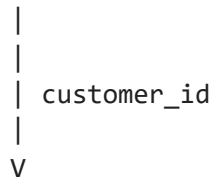
```
Out[23]: 99441
```

```
In [24]: orders['order_id'].nunique()
```

```
Out[24]: 99441
```

CUSTOMERS

PK customer_id customer_name city



ORDERS

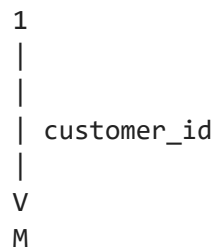
PK order_id FK customer_id order_date

One Customer ↓ Many Orders

CUSTOMERS

PK customer_id

customer_unique_id customer_city customer_state



ORDERS

PK order_id

FK customer_id

order_status order_purchase_timestamp order_approved_at order_delivered_carrier_date
order_delivered_customer_date order_estimated_delivery_date

In [25]: `import os`

`os.listdir()`

Out[25]: `['.ipynb_checkpoints',
'healthcare.db',
'healthcare_cost_analytics.ipynb',
'insurance.csv',
'olist.duckdb',
'Olist_SQL_Project',
'Project_1_HealthcareAnalytics.ipynb',
'Project_2_Data_Quality_Assessment.ipynb',
'Project_2_Data_Understanding.ipynb.ipynb']`

In [26]: `for file in os.listdir("Olist_SQL_Project/data"):
 print(file)`

`.ipynb_checkpoints
olist_customers_dataset.csv
olist_geolocation_dataset.csv
olist_orders_dataset.csv
olist_order_items_dataset.csv
olist_order_payments_dataset.csv
olist_order_reviews_dataset.csv
olist_products_dataset.csv
olist_sellers_dataset.csv
product_category_name_translation.csv`

In [27]: `order_items.head()`

Out[27]:

	order_id	order_item_id	product_id
0	00010242fe8c5a6d1ba2dd792cb16214	1	4244733e06e7ecb4970a6e2683c13e61
1	00018f77f2f0320c557190d7a144bdd3	1	e5f2d52b802189ee658865ca93d83a8f
2	000229ec398224ef6ca0657da4fc703e	1	c777355d18b72b67abbeef9df44fd0fd
3	00024acbcdf0a6daa1e931b038114c75	1	7634da152a4610f1595efa32f14722fc
4	00042b26cf59d7ce69dfabb4e55b4fd9	1	ac6c3623068f30de03045865e4e10089

◀  ▶

In [28]: `order_items.shape`

Out[28]: `(112650, 7)`

In [29]: `order_items.columns`

Out[29]: `Index(['order_id', 'order_item_id', 'product_id', 'seller_id',
 'shipping_limit_date', 'price', 'freight_value'],
 dtype='str')`

```
In [30]: order_items.info()
```

```
<class 'pandas.DataFrame'>  
RangeIndex: 112650 entries, 0 to 112649  
Data columns (total 7 columns):  
#   Column                Non-Null Count  Dtype  
---  ---  
0   order_id              112650 non-null  str  
1   order_item_id         112650 non-null  int64  
2   product_id            112650 non-null  str  
3   seller_id             112650 non-null  str  
4   shipping_limit_date   112650 non-null  str  
5   price                 112650 non-null  float64  
6   freight_value         112650 non-null  float64  
dtypes: float64(2), int64(1), str(4)  
memory usage: 6.0 MB
```

```
In [31]: order_items[['order_id', 'order_item_id']].drop_duplicates().shape[0]
```

```
Out[31]: 112650
```

```
In [32]: order_items['order_id'].isin(  
         orders['order_id']  
         ).all()
```

```
Out[32]: np.True_
```

```
In [33]: orders['order_id'].nunique() == len(orders)
```

```
Out[33]: True
```

```
In [34]: orders['order_id'].isnull().sum()
```

```
Out[34]: np.int64(0)
```

```
In [35]: order_items['product_id'].isin(  
         products['product_id']  
         ).all()
```

```
Out[35]: np.True_
```

```
In [36]: products['product_id'].nunique() == len(products)
```

```
Out[36]: True
```

```
In [37]: products['product_id'].isnull().sum()
```

```
Out[37]: np.int64(0)
```

```
In [38]: order_items['seller_id'].isin(  
         sellers['seller_id']  
         ).all()
```

Out[38]: np.True_

```
In [39]: sellers['seller_id'].nunique() == len(sellers)
```

Out[39]: True

```
In [40]: sellers['seller_id'].isnull().sum()
```

Out[40]: np.int64(0)

```
In [41]: translation['product_category_name'].isin(
    products['product_category_name']
).all()
```

Out[41]: np.True_

```
In [42]: translation['product_category_name'].dropna().isin(
    products['product_category_name']
).all()
```

Out[42]: np.True_

```
In [43]: products.loc[
    ~products['product_category_name'].isin(
        translation['product_category_name']
    ),
    'product_category_name'
].unique()
```

Out[43]: <StringArray>
[nan, 'pc_gamer', 'portateis_cozinha_e_preparadores_de_alimentos']
Length: 3, dtype: str

```
In [45]: order_items[['order_id', 'order_item_id']].drop_duplicates().shape[0]
```

Out[45]: 112650

```
In [46]: len(order_items)
```

Out[46]: 112650

Composite PK = (order_id, order_item_id)

Customers PK - customer_id FK - NA Orders PK - order_id FK - customer_id Order_items Composite PK = (order_id, order_item_id) FK - order_id, seller_id, product_id products PK - product_id FK -

```
In [47]: products.shape
```

Out[47]: (32951, 9)

```
In [48]: products['product_id'].shape[0]
```

Out[48]: 32951

```
In [49]: products['product_id'].nunique()
```

```
Out[49]: 32951
```

```
In [50]: products['product_id'].isnull().sum()
```

```
Out[50]: np.int64(0)
```

```
In [51]: for col in products.columns:
          print(col, products[col].nunique(), len(products))
```

```
product_id 32951 32951
product_category_name 73 32951
product_name_lenght 66 32951
product_description_lenght 2960 32951
product_photos_qty 19 32951
product_weight_g 2204 32951
product_length_cm 99 32951
product_height_cm 102 32951
product_width_cm 95 32951
```

```
In [52]: sellers.sample(5)
```

```
Out[52]:
```

	seller_id	seller_zip_code_prefix	seller_city	seller_state
2552	57fe6232e67823593374a2281d7a8af8	13360	capivari	SP
230	8ca5b19e0263150d2c57afdde2b3d011	9336	maua	SP
1463	7c462184585ae80bd6e5774c62ce50f2	13086	campinas	SP
1470	e2a1ac9bf33e5549a2a4f834e70df2f8	1210	sao paulo	SP
2785	a1474fd46fd3430168f7ec202da39408	83321	pinhais	SP

```
In [53]: sellers['seller_id'].nunique() == len(sellers)
```

```
Out[53]: True
```

```
In [54]: sellers['seller_id'].drop_duplicates().shape[0] == len(sellers)
```

```
Out[54]: True
```

```
In [55]: sellers['seller_id'].isnull().sum()
```

```
Out[55]: np.int64(0)
```

```
In [56]: payments.sample(5)
```

Out[56]:

	order_id	payment_sequential	payment_type	payment_ir
67864	6c066ac65ae3a217178d6e90f682be03	1	credit_card	
8369	206180d63bbc3cde4e902195b8493bad	1	boleto	
98108	ac28767a818e6e5bc6895e188ef622ba	1	credit_card	
74657	0fcb1b4cfdb2f127f9ae4d732352dfc7	1	boleto	
49792	5ad066b6e468fcd60b0d871bfa99112a	1	credit_card	



In [57]: `len(payments)`

Out[57]: 103886

In [58]: `payments[['order_id', 'payment_sequential']].drop_duplicates().shape[0]`

Out[58]: 103886

In [59]: `payments[['order_id', 'payment_sequential']].isnull().sum()`

Out[59]:

order_id	0
payment_sequential	0

dtype: int64

In [60]: `import duckdb`

`con = duckdb.connect("olist.duckdb")`

`con.sql("""`
`CREATE OR REPLACE TABLE customers AS`
`SELECT *`
`FROM read_csv_auto('Olist_SQL_Project/data/olist_customers_dataset.csv');`
`""")`

`con.sql("""`
`CREATE OR REPLACE TABLE orders AS`
`SELECT *`
`FROM read_csv_auto('Olist_SQL_Project/data/olist_orders_dataset.csv');`
`""")`

`con.sql("""`
`CREATE OR REPLACE TABLE order_items AS`
`SELECT *`
`FROM read_csv_auto('Olist_SQL_Project/data/olist_order_items_dataset.csv');`
`""")`

`con.sql("""`
`CREATE OR REPLACE TABLE products AS`
`SELECT *`
`FROM read_csv_auto('Olist_SQL_Project/data/olist_products_dataset.csv');`
`""")`

```

con.sql("""
CREATE OR REPLACE TABLE sellers AS
SELECT *
FROM read_csv_auto('Olist_SQL_Project/data/olist_sellers_dataset.csv');
""")

con.sql("""
CREATE OR REPLACE TABLE payments AS
SELECT *
FROM read_csv_auto('Olist_SQL_Project/data/olist_order_payments_dataset.csv');
""")

con.sql("""
CREATE OR REPLACE TABLE reviews AS
SELECT *
FROM read_csv_auto('Olist_SQL_Project/data/olist_order_reviews_dataset.csv');
""")

con.sql("""
CREATE OR REPLACE TABLE translation AS
SELECT *
FROM read_csv_auto('Olist_SQL_Project/data/product_category_name_translation.csv');
""")

con.sql("""
CREATE OR REPLACE TABLE geolocation AS
SELECT *
FROM read_csv_auto('Olist_SQL_Project/data/olist_geolocation_dataset.csv');
""")

print("Tables Created Successfully!")

```

Tables Created Successfully!

In [61]: `con.sql("SHOW TABLES").show()`

name varchar
customers
geolocation
order_items
orders
payments
products
reviews
sellers
translation

In [64]: `con.sql("""
SELECT 'customers' AS table_name, COUNT(*) AS rows FROM customers
UNION ALL
SELECT 'orders', COUNT(*) FROM orders
UNION ALL`

```

SELECT 'order_items', COUNT(*) FROM order_items
UNION ALL
SELECT 'products', COUNT(*) FROM products
UNION ALL
SELECT 'sellers', COUNT(*) FROM sellers
UNION ALL
SELECT 'payments', COUNT(*) FROM payments
UNION ALL
SELECT 'reviews', COUNT(*) FROM reviews
UNION ALL
SELECT 'translation', COUNT(*) FROM translation
UNION ALL
SELECT 'geolocation', COUNT(*) FROM geolocation
""").show()

```

table_name varchar	rows int64
customers	99441
orders	99441
order_items	112650
products	32951
sellers	3095
payments	103886
reviews	99224
translation	71
geolocation	1000163

```
In [65]: con.close()
```

```
In [ ]:
```